Machine Learning and Theorem Proving

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Outline

Motivation, Learning vs. Reasoning

Motivation: Learning vs. Reasoning

"C'est par la logique qu'on démontre, c'est par l'intuition qu'on invente."
(It is by logic that we prove, but by intuition that we discover.)
Henri Poincaré, Mathematical Definitions and Education.

"Hypothesen sind Netze; nur der fängt, wer auswirft."
(Hypotheses are nets: only he who casts will catch.)
Novalis, quoted by Popper – The Logic of Scientific Discovery

Leibniz's/Hilbert's/Russell's Dream: Let Us Calculate!

Solve all (math, physics, law, economics, society, ...) problems by
reduction to logic/computation

[Adapted from: *Logicomix: An Epic Search for Truth* by A. Doxiadis]